PIONEER MAIZE INSECT PEST SERIES – COSMOPOLITAN ARMYWORM

Pioneer® Technical Insight 329

INTRODUCTION

Cosmopolitan Armyworm (Leucania (=Pseudaletia) separate) is found throughout the North Island and Upper South Island but is most common in South Auckland and Northland. The name armyworm comes from the marching habit of large numbers of larvae which, once they have depleted their food supply, move in large groups in search of more food. Although it was once a major pest in maize in the northern regions of the country, Cosmopolitan Armyworm seldom causes economic damage in maize since the introduction of a biological control – a parasitic wasp Apantheles ruficrus.

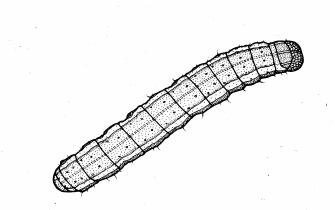
ARMYWORM DESCRIPTION

Adult armyworms are a dull yellowish to reddish brown moth with a wingspan of 40-50 mm. They have a very small central whitish spot on the forewings. Adults are nocturnal and are strongly attracted to light.

Eggs are pale cream in colour. They are normally laid under the leaf sheaths of the maize or grass plants.

Caterpillars are up to 40 mm when fully grown. The younger caterpillars are green whereas the larger caterpillars are pale greenish brown with parallel light and dark lines down the length of the body. The head has a honeycomb pattern on it (Figure 1).

Figure 1: Cosmopolitan Armyworm Caterpillar



Pupae are reddish brown and are found underground or beneath stones.

LIFE CYCLE

Cosmopolitan Armyworms have three to four generations per year in most parts of New Zealand. The generation time is 6-8 weeks.

The spring flight of moths results from overwintering larvae. Eggs are laid under leaf sheaths. Caterpillar density reaches a maximum in February to April. Each caterpillar passes through six to eight growth stages (instars). 85% percent of plant consumption occurs in the last two instars and typically the lower part of the plant is consumed before the caterpillar moves up to the upper parts of the plant. The development of all stages slows as the weather cools. All stages overwinter in reduced numbers.

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MAIZE DAMAGE

Maize damage is caused by the larvae feeding from the edge towards the centre of the leaves. They chew irregular chunks giving a saw-like appearance and under severe damage, only the leaf midrib remains. Damage is more prevalent where there is grass weed cover at the base of the crop. The caterpillars normally start on the lower leaves and move up the plant.

Crops should be monitored from New Year onwards if infestation is thought likely.

CONTROL OPTIONS

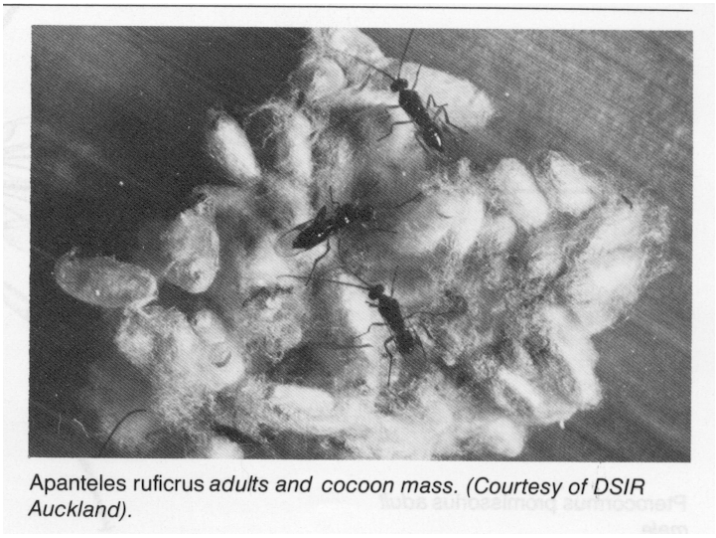
Effective Pre Emergent Grass Weed Control

Ensure that there is an effective pre emergent grass weed control program in place. Crops with good grass weed control seldom suffer economic damage from Cosmopolitan Armyworm infestation.

Biological Control

The main form of control of Cosmopolitan Armyworm in New Zealand is an introduced strain of the parasitic wasp *Apanteles ruficrus*. Introduced from Pakistan in 1971-72, *Apanteles ruficrus* is a small black-brown wasp around 4 mm in length. It lays its eggs within the young armyworm larvae. The armyworm continues to develop and feed normally with the developing parasite larvae inside it. When the armyworm caterpillar is around 20 mm long, the *Apanteles ruficrus* larvae chew through its skin and emerge to spin pale white or yellowish silken cocoons (Fig 2).

Figure 2: *Apanteles ruficrus* adults and cocoons



Apanteles ruficrus *adults and cocoon mass. (Courtesy of DSIR Auckland).*

Clusters of cocoons are usually seen on the cob or at the base of the leaves (near the stem). The armyworm then crawls away and dies. Adult wasps emerge after 10-14 days and begin searching for new armyworm caterpillars to lay their eggs in. Normally around 60 larvae are laid in each caterpillar.

Apantheles ruficrus infests up to 98% of the caterpillars in some crops. The armyworm caterpillars die just before they enter the stage that causes 80% of crop damage. In most cases armyworm caterpillars are kept well below economic injury level.

To check for *Apantheles ruficrus* parasitism of armyworm caterpillars, choose caterpillars that are 10-20 mm in length and break them in half by pulling either end. The wasp larvae can be clearly seen amongst the gut contents of the armyworm.

Occasionally high levels of caterpillars that have not been parasitised are found and it may be necessary to consider spraying the crop.

When making the decision to spray or not to spray, consider the fact the chemical will also kill the parasitic wasp population within the crop area.

CHEMICAL CONTROL

Chemical control of armyworm in maize crops should only be considered if the number of caterpillars is high (more than 10 per plant) and the level of wasp parasitism is low.

There are a range of insecticides registered for the control of Cosmopolitan Armyworm in maize. These include: Carbaryl, Chlorpyrifos, Alpha-Cypermethrin, Methamidophos, Ripcord® and Sumi-Alpha®.

Spray application by air is the only option since crop height rules out ground application. It is critical that the water application rate is high enough to ensure that the insecticide reaches the base of the maize plant where the bulk of the larvae feed.

REFERENCES

Cosmopolitan army caterpillar. DSIR Information Series No 105/14

New Zealand Pests and Beneficial Insects (1984) Scott, R.R (Ed) Lincoln University

Maize- management to market. Agronomy Society of New Zealand. Special Publication No. 4

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